Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Final Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ Grade & Sec: Grade10 _____ R.No: _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** When balancing an equation, you can change the subscripts within a chemical formula to make atoms equal.

____ **2.** A substance that gets oxidized acts as the oxidizing agent and reduced acts as reducing agent.

____ **3.** A chemical reaction always produces a new substance with different properties.

____ **4.** The solubility of a solid in liquid decreases if the dissolution process is endothermic.

____ **5.** When a solution is diluted by adding more solvent the number of moles of solute in solution is changed.

____ **6.** The rate of dissolution of a solid solute generally increases when the surface area of solute is increased.

____ **7.** Pressure has a significant effect on the solubility of a solid solute in liquid solvent.

II. Match the items under Column A with the correct descriptions under Column B.

8. Match the reaction description with its classification: "Forms a single product from two or more reactants."

A. Combination reaction
B. Redox reaction
C. Decomposition reaction
D. Single displacement reaction
E. Double displacement reaction

9. Match the reaction description with its classification: "Breaking down a single compound into two or more elements or simpler compounds."

A. Combination reaction
B. Redox reaction
C. Decomposition reaction
D. Single displacement reaction
E. Double displacement reaction

10. Match the reaction description with its classification: "Replacement reaction where an uncombined element displaces another element from a compound."

A. Combination reaction
B. Redox reaction
C. Decomposition reaction
D. Single displacement reaction
E. Double displacement reaction

11. Match the reaction description with its classification: "Exchange of positive and negative ions between two compounds."

A. Combination reaction
B. Redox reaction
C. Decomposition reaction
D. Single displacement reaction
E. Double displacement reaction

12. Match the reaction description with its classification: "The transfer of electrons between chemical species."

A. Combination reaction
B. Redox reaction
C. Decomposition reaction
D. Single displacement reaction
E. Double displacement reaction

III. Choose the best answer among the given alternatives.

____ **13.** A 0.5L and 0.1M HNO_3 solution is to be prepared by dilution from 4M HNO_3. How many mL of concentrated HNO_3 and how many mL of water are required to prepare the dilution?

A. 6.5 mL HNO_3 and 480 mL water
B. 12.5 mL HNO_3 and 487.5 mL water
C. 10 mL HNO_3 and 400 mL water
D. 8 mL HNO_3 and 480.5 mL water

____ **14.** A solution of NH_4Cl made by dissolving 5.4g NH_4Cl in 36.01g H_2O has a density of 1.027 g/mL. What is the mole fraction of NH_4Cl? (Atomic masses: Cl=35.5 ~ 36, N=14, H=1)

A. 0.05 **B.** 0.476 **C.** 0.0476 **D.** 0.051

____ **15.** Which of the following explains dipole-dipole forces?

A. The intermolecular attraction force that exists between non-polar molecules

B. A type of intermolecular force that exists between polar molecules
C. An electrostatic attraction between metal ions and valence electrons
D. An attractive force resulting from temporary dipoles induced in molecules

____ **16.** The partial pressure of acetylene, C_2H_2, in a container is 10 atm at 25°C. What is the solubility in molarity of C_2H_2, assuming C_2H_2 obeys Henry's Law? (K_H = 1.04 mol/L·atm)

A. 0.104 mol/L **B.** 9.62 mol/L **C.** 10.4 mol/L **D.** 104 mol/L

____ **17.** What is the normality of a solution that contains 19.6 g of H_2SO_4 in 20 mL of solution?

A. 0.01 N **B.** 0.2 N **C.** 20 N **D.** 10 N

____ **18.** The difference between colloids and suspensions is that:

A. Colloidal particles are small enough and do not settle down unlike in a suspension
B. Most colloids and suspensions appear cloudy
C. Both colloids and suspensions are mixtures
D. Colloids are as transparent as solutions

____ **19.** What is the molarity of 3.4g of hydrogen peroxide (H_2O_2) in 200 mL of solution used for hair bleaching? (Molar mass H_2O_2 = 34 g/mol)

A. 0.05 M **B.** 0.5 M **C.** 0.15 M **D.** 0.015 M

____ **20.** How many moles of $AgNO_3$ are needed to react completely with 50 mL of 0.2M $MgCl_2$? ($2AgNO_3 + MgCl_2 \rightarrow 2AgCl + Mg(NO_3)_2$)

A. 0.04 mol **B.** 0.01 mol **C.** 0.005 mol **D.** 0.02 mol

____ **21.** What volume of 0.5M H_2SO_4 is required to neutralize 25 mL of 1M KOH?

A. 50 mL **B.** 12.5 mL **C.** 25 mL **D.** 100 mL

____ **22.** How many grams of $CaCO_3$ are produced from 500 mL of 0.1M Na_2CO_3 reacting with excess $CaCl_2$? ($Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + 2NaCl$; Atomic masses: Ca=40, C=12, O=16)

A. 5 g **B.** 2 g **C.** 50 g **D.** 10 g

____ **23.** When $BaCl_2(aq)$ and $Na_2SO_4(aq)$ are mixed, a white precipitate forms. What is the identity of this precipitate?

A. NaCl **B.** $BaSO_4$ **C.** $BaCl_2$ **D.** $NaSO_4$

____ **24.** If 40g of sodium hydroxide (NaOH) is dissolved in 2000g of water, what is the molality of the solution? (Molar mass NaOH = 40 g/mol)

A. 1.0 m **B.** 2.0 m **C.** 0.25 m **D.** 0.5 m

____ **25.** What is the percent by mass of sugar in a solution prepared by dissolving 10 g of sugar into a total solution mass of 200 g?

A. 5 % **B.** 0.5 % **C.** 0.25 % **D.** 50 %

____ **26.** Which of the following chemical reactions is NOT balanced?

A. 2NaHCO3 -> Na2CO3 + H2O + CO2
B. 2C4H10 + 12O2 -> 8CO2 + 10H2O
C. 2Al + 6H2O -> 2Al(OH)3 + 3H2
D. 4NH3 + 5O2 -> 4NO + 6H2O

____ **27.** Which of the following single replacement reactions is NOT correct and will not occur as written?

A. Zn + CuSO4 -> ZnSO4 + Cu
B. 2Ag + Cu(NO3)2 -> AgNO3 + Cu
C. Fe + CuSO4 -> FeSO4 + Cu
D. Mg + 2HCl -> MgCl2 + H2

____ **28.** What volume of a 0.5M sugar solution contains 3.011 * 10^22 molecules of sugar?

A. 0.01 L **B.** 0.1 L **C.** 0.5 L **D.** 0.05 L

____ **29.** For the chemical reaction Na2SO4(aq) + Ba(NO3)2(aq) -> BaSO4(s) + 2NaNO3(aq), what are the spectator ions?

A. Na+ and NO3- **B.** SO4(2-) and NO3- **C.** Na+ and Ba2+ **D.** Ba2+ and SO4(2-)

____ **30.** Which one of the following molecules does NOT form hydrogen bonding among its molecules?

A. HF **B.** H2O **C.** NH3 **D.** PCl5

____ **31.** All of the following compounds are soluble in water except one:

A. AgNO3 **B.** CH3COONa **C.** CaCO3 **D.** K3PO4

____ **32.** The rate of dissolution is fundamentally dependent upon:

A. The inter-particle force
B. The surface area
C. The temperature of the system
D. The pressure of the system

____ **33.** To prepare 250 mL of a 10% (V/V) vinegar solution, how many milliliters of pure acetic acid solute are needed?

A. 2.5 mL **B.** 25 mL **C.** 250 mL **D.** 100 mL